

Ranking H1 through H6

The review does not generate hypotheses. Six already exist, drawn from a human peer-play transcript. Its job is to rank them and decide which one gets built first.

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Status screening and extraction complete **Ranking** not yet written

Why the review exists

A prior transcript analysis extracted 30 cognitive abstractions from watching a human play, then mapped them onto 48 research topics in five clusters. It closed with an explicit instruction, and this review is the response to it.

“I would stop here before designing hypotheses. The transcript has given you a vocabulary of mechanisms; the next stage should be literature review to determine which of these mechanisms already have strong formalizations, which overlap, and which are genuinely underexplored in ARC-like interactive settings.”

TRANSCRIPT ANALYSIS, TAB 3

Framing it as a ranking task is what makes it finite. Without a decision to serve, a citation-graph review expands indefinitely — there is always one more hop.

The question the review must answer

Of H1–H6, which mechanism is **underexplored in interactive settings, implementable without training**, and **able to run offline on a Kaggle GPU** — and therefore the right first

modification to the baseline agent?

Those three conditions are not preferences. Each is forced by something outside the research question:

CONSTRAINT 01	CONSTRAINT 02	CONSTRAINT 03
Underexplored, interactively The stated aim of the transcript analysis — and it stops us rebuilding work that already exists under another name.	No training Fine-tuning, RL and imitation learning are all out of scope for this milestone by the project's own initialisation rules.	Runs offline Kaggle evaluation is sandboxed with no internet. No hosted-API model can be called at evaluation time.

Constraint 03 is the one that reorders everything. It is also, as the extraction later showed, the constraint that makes the research question answerable at all — mechanism differences are masked at frontier scale and only become measurable under a compute bound.

The five clusters

The filing system for everything the review collects. Defined before any paper was read, and used to keep the shortlist balanced — a review returning twenty exploration papers and nothing on memory cannot rank these hypotheses.

A

Understanding the unknown world

What exists? What can I influence? What happens when I act?

System identification · causal discovery through interventions · affordance learning · object-centric world models · factored state · relational reasoning · controllability discovery · temporal dynamics

9 papers in the final shortlist · owns H3, H4

B

Deciding what is probably true

What do I know versus merely suspect, and how should evidence move that?

Bayesian inference and belief states · hypothesis generation and testing · epistemic uncertainty · program induction · representation revision · inductive biases and priors · compositional generalisation

9 papers · owns H1

C

Discovering useful information

Which action should I take when the point of the action is to learn?

Active learning · Bayesian experimental design · value of information · intrinsic motivation · curiosity-driven exploration · directed and novelty-based exploration · exploration–exploitation

15 papers · owns H2, H5

D

Turning knowledge into action

Given my current model, what should I actually do?

Model-based RL · hierarchical planning · subgoal discovery · receding-horizon control · tree search · width-based planning · POMDPs · resource-constrained and risk-sensitive planning · plan repair

4 papers · deliberately light – this is Experiment 2 territory

E

Improving over experience

What should remain after this action, episode, level, or game?

Working, episodic and semantic memory · consolidation · selective forgetting · meta-learning · continual learning · learning from failure · metacognition · bounded rationality · cognitive maps · skill chunking

21 papers · owns H6 · the deepest recent literature

Cluster D is under-weighted on purpose. Experiment 1 tests whether structured memory helps, not whether planning helps. Adding a planner would confound the comparison, so D gets a small quota by design rather than by neglect.

The six hypotheses

Each is an ablation over the *same* base agent, so they stay independently measurable. Every one traces to something a human actually did in the transcript — that is where

they came from, and it is why they are worth testing rather than merely plausible.

H1

CLUSTER B

Tagging memory as **fact versus hypothesis**, with a confidence level, beats a flat observation log.

"I think..." shifting to "I'm very sure..."

CLAIMED ON ARC-AGI-3 · 14 PAPERS

H2

CLUSTER C

Labelling each action **epistemic versus instrumental**, under an exploration budget, improves efficiency.

Touching the plus to learn what it does, later to use it

CLAIMED AND NAMED · 14 PAPERS

H3

CLUSTER A

Spending the first **k** actions identifying the **controllable object** lowers total actions.

"What can I control?" as the opening move

UNCLAIMED ON ARC-AGI-3 · 6 PAPERS

H4

CLUSTER A

Separating **self-caused from world-caused** change reduces false causal attribution.

The oscillating plus sign moving on its own

CLAIMED ON ARC-AGI-3 · 8 PAPERS

H5

CLUSTER C + D

Pruning actions that **return to visited states** reduces redundancy.

Going left "unwraps" a new state; right returns

UNCLAIMED ON ARC-AGI-3 · 7 PAPERS

H6

CLUSTER E

Promoting **episodic failures into semantic rules** prevents repeat deaths across resets.

"I know the flame kills me because I failed"

CONTESTED · 18 PAPERS

H3 and H5 are the only two nobody has claimed on this benchmark — and they are also the two with the thinnest support in our own shortlist. That is either open ground or an absence of evidence, and the difference matters enough to settle by reading rather than by novelty.

A live threat to H3: Tycho reports that high transition accuracy *did not* improve action selection. A better world model is not automatically a better agent.

Seeds, and what they could not reach

A seed is a paper already trusted, used as an entry point into the citation graph. They were chosen for *coverage* — together they must open a path into all five clusters — not for individual quality.

The plan called for six recent seeds and a backward-only hop, on the reasoning that 2026 papers have near-empty forward citations. Both halves of that were revised. Five classical seeds were added to reach the mechanism literature the recent work assumes but does not cite, and forward hops were run from each, giving **11 seeds and 8 forward lists**.

The blind spot. Thirteen ARC-AGI-3 papers exist. 512 backward references and 8 forward lists surfaced *none* of them, because every seed either predates the benchmark or is contemporary with it, and none cite it.

A citation graph cannot find a literature that postdates its seeds. The fix costs one extra search — always pair the graph walk with a direct name search for the benchmark, the method and the competition.

Planned versus actual

The plan is kept alongside what happened rather than rewritten to match it. Three rules in this plan were overturned by their own output, and a review that hides its corrections is not reproducible.

DECISION	PLANNED	ACTUAL	WHY IT MOVED
Seeds	6	11	The 2026 seeds do not cite the mechanism literature they assume.
Hop direction	backward only	both	Forward hops were not empty after all.
Shortlist size	25	58	Widened three times — for the classical seeds, then for the ARC-AGI-3 papers, then to restore displaced work.

DECISION	PLANNED	ACTUAL	WHY IT MOVED
Ranking signal	citation count	cluster-keyword density	Fame is not relevance. Citation ranking returned <i>Matplotlib</i> as a world-models paper.
Cluster assignment	during triage	after extraction	With the mechanism visible, 7 of 58 assignments changed.
Extraction columns	9 planned	9 shipped	Three replaced: cluster is our judgement not the paper's; training and code folded into model/limitations.

The review ran roughly a day behind plan and one scope-step wider, both for the same reason: the ARC-AGI-3 literature discovered mid-review did not exist in the citation graph and had to be found, verified and triaged by hand.

Where the plan stands

CLOSED	Seed set confirmed and expanded to 11; both unread seeds resolved.
CLOSED	Extraction workflow chosen and run — Extract Data on Elicit Pro, two tables, nine columns.
CLOSED	<code>cluster_final</code> filled for all 58, from the extraction rather than ahead of it.
BLOCKING	Kaggle account and phone verification. Gates GPU access, still outstanding, and the only item with an external dependency.
OPEN	The ranking write-up. The deliverable this plan exists to produce — H1–H6 ranked with evidence, plus a prior-art note so the build does not rebuild Tycho or AERA by accident.
OPEN	The model decision. Narrowed but not settled: a submission forces local weights, and 13 shortlisted papers show small open-weight models working in adjacent settings. No model named.

OPEN

Two unreachable papers. Excluded by decision, not by screening. Coverage is 56 of 58, and the write-up must state that plainly.

Companion notes: *From 590 to 58* for how the set was assembled, and *What the 58 Say* for what the extraction found. Method is citation-graph snowballing with hand triage — **not PRISMA**. Current to 5 September 2026.